



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester II Examinations 2023–2024

Course Instance Codes(s) 3BA1, 4BME1, 4BMU1, 1EM1, 4BCT1, 3BMS2, 4BMS2, 3BS9
Exam(s) 3rd Arts, 4th Arts, 3rd Science,

3rd BA Education, Masters.

Module Codes MA3491

Module Fields and Applications

Paper No 1

External Examiner Prof. C. Roney-Dougal

Internal Examiner(s) Dr N. Madden

★ Dr E. Sköldberg

Instructions:

Answer all FIVE questions

Duration 2 Hours

Number of Pages 3 (including this page)

Discipline Mathematics

Requirements:

Release in Exam Venue Yes ☒ No ☐

Release to Library Yes ☒ No ☐

Statistical/ Log Tables: State Examination Commission's *Formulae and Tables*

Handout None

Cambridge Tables None

Graph Paper None

Log Graph Paper None

Other Materials None

Graphic material in colour Yes ☐ No ☒

- Q1. (a) State the *sphere packing bound*, and determine the upper bound it gives on the number of code words in a ternary code with block length 8 and minimum distance 3. [6 marks]
- (b) State the *Singleton bound*, and determine the upper bound it gives on the number of code words in a 4-ary code with block length 8 and minimum distance 3. [6 marks]
- (c) Let $P_{n,q}$ be the block code over an alphabet of size q , and with block length n , consisting of all *palindromic* words, i.e. words that read the same forwards and backwards.
- (i) Let the blocklength $n = 2k$ be even. Determine the *number of codewords* and the *minimum distance* of $P_{2k,q}$. [4 marks]
- (ii) Let the blocklength $n = 2k + 1$ be odd. Determine the *number of codewords* and the *minimum distance* of $P_{2k+1,q}$. [4 marks]
- Q2. (a) Factor the polynomial $x^6 - 1 \in \mathbb{F}_3[x]$ into irreducibles. [6 marks]
- (b) (i) Show that $f(x) = x^4 + x + 1 \in \mathbb{F}_2[x]$ is irreducible, and hence let $\mathbb{F}_{16} = \mathbb{F}_2[x]/(f(x))$. [3 marks]
- (ii) Find the order of the element $x^2 + x$ in $\mathbb{F}_{16}$. [5 marks]
- (iii) Calculate the multiplicative inverse of x^2 in $\mathbb{F}_{16}$. (Give the result as a polynomial of degree ≤ 3). [6 marks]

- Q3. (a) Let C_1 be the linear ternary code given by the generator matrix

$$G = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

- (i) Write down the code words of C_1 . [4 marks]
- (ii) What is the minimum distance of C_1 ? [2 marks]
- (iii) Construct a *standard array* for C_1 , and use that to decode the received words 122 and 111. [6 marks]
- (b) Consider the linear code C_2 over $\mathbb{F}_3$ with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}.$$

- (i) Construct a check matrix for C_2 . [2 marks]
- (ii) Construct a syndrome table for C_2 and use it to decode the word 1111. [6 marks]

- Q4. (a) Let C_3 be the linear $[4, 1]$ -code over $\mathbb{F}_7$ with check matrix

$$\begin{pmatrix} 0 & 0 & 1 & 3 \\ 1 & 1 & 1 & 4 \\ 0 & 1 & 1 & 5 \end{pmatrix}.$$

Determine the minimum distance of C_3 . [6 marks]

- (b) (i) What is the definition of the Hamming code $Ham(r, q)$? [4 marks]
- (ii) Construct a parity check matrix for the Hamming code $Ham(2, 4)$. [4 marks]
- (iii) Explain how a single error can be corrected in $Ham(2, 4)$. Decode the words 11111, and 01010. [6 marks]

- Q5. (a) (i) Define the *weight enumerator* of an $[n, k]$ -code. [2 marks]
(ii) Compute the weight enumerator of the ternary $[4, 2]$ -code given by the generator matrix

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 0 \end{pmatrix}.$$

[6 marks]

- (b) Let C_4 be the cyclic code of length 6 over the field $\mathbb{F}_3$ with generator polynomial $x^4 + 2x^3 + x + 2$.
(i) Construct a generator matrix for C_4 . [4 marks]
(ii) Find the check polynomial for C_4 and hence construct a check matrix for C_4 . [8 marks]